

Salvatore D. Pace

Curriculum Vitae

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Academic Positions

Institute for Advanced Study

Princeton, NJ

Member, School of Natural Sciences

Start date: Sept 2026

Education

Massachusetts Institute of Technology

Cambridge, MA

Ph.D. in Physics

2026

Thesis: *When Symmetry Does the Unexpected*

Advisor: Xiao-Gang Wen

University of Cambridge

Cambridge, UK

MPhil in Physics

2021

Thesis: *Emergent Axions in $U(1)$ Quantum Spin Liquids*

Advisor: Claudio Castelnovo

Boston University

Boston, MA

B.A. with honors & M.A. in Physics

2020

Thesis: *The Fine Structure Constant in Quantum Spin Ice*

Advisor: Christopher Laumann

Selected Awards and Honors

MIT Andrew M. Lockett Award for Doctoral Theoretical Physics Research

2026

Kavli Institute for Theoretical Physics Graduate Fellowship

2025

National Science Foundation Graduate Research Fellowship

2022 – 2025

Churchill Scholarship

2020 – 2021

American Physical Society LeRoy Apker Award Finalist

2020

BU College Prize for Excellence in the Physics Department

2020

Learning Assistant of the Year

2019

Goldwater Scholarship

2019

Scientific Papers [[Google Scholar](#)]

[19] [S. D. Pace](#) and D. Bulmash, “Lieb-Schultz-Mattis constraints from stratified anomalies of modulated symmetries,” [arXiv:2602.11266](#).

[18] M. L. Kim, [S. D. Pace](#), and S.-H. Shao, “Symmetry-enforced Fermi surfaces,” *Phys. Rev. Lett.* **136** (2026) 176502, [[arXiv:2512.04150](#)].

- [17] [S. D. Pace](#), [Ö. M. Aksoy](#), and [H. T. Lam](#), “Spacetime symmetry-enriched SymTFT: from LSM anomalies to modulated symmetries and beyond,” *SciPost Phys.* **20** (2026) 007, [[arXiv:2507.02036](#)].
- [16] [S. D. Pace](#), [M. L. Kim](#), [A. Chatterjee](#), and [S.-H. Shao](#), ‘Parity Anomaly from a Lieb-Schultz-Mattis Theorem: Exact Valley Symmetries on the Lattice,” *Phys. Rev. Lett.* **135** (2025) 236501, [[arXiv:2505.04684](#)].
- [15] [S. D. Pace](#), [A. Chatterjee](#), and [S.-H. Shao](#), “Lattice T-duality from non-invertible symmetries in quantum spin chains,” *SciPost Phys.* **18** (2025) 121, [[arXiv:2412.18606](#)].
- [14] [S. D. Pace](#), [H. T. Lam](#), and [Ö. M. Aksoy](#), “(SPT-)LSM theorems from projective non-invertible symmetries,” *SciPost Phys.* **18** (2025) 028, [[arXiv:2409.18113](#)].
- [13] [A. Chatterjee](#), [S. D. Pace](#), and [S.-H. Shao](#), “Quantized axial charge of staggered fermions and the chiral anomaly,” *Phys. Rev. Lett.* **134** (2025) 021601, [[arXiv:2409.12220](#)].
- [12] [S. D. Pace](#), [G. Delfino](#), [H. T. Lam](#), and [Ö. M. Aksoy](#), “Gauging modulated symmetries: Kramers-Wannier dualities and non-invertible reflections,” *SciPost Phys.* **18** (2025) 021, [[arXiv:2406.12962](#)].
- [11] [S. D. Pace](#) and [Y. L. Liu](#), “Topological aspects of brane fields: Solitons and higher-form symmetries,” *SciPost Phys.* **16** (2024) 128, [[arXiv:2311.09293](#)].
- [10] [S. D. Pace](#), [C. Zhu](#), [A. Beaudry](#), and [X.-G. Wen](#), “Generalized symmetries in singularity-free nonlinear σ models and their disordered phases,” *Phys. Rev. B* **110** (2024) 195149, [[arXiv:2310.08554](#)].
- [9] [S. D. Pace](#), “Emergent generalized symmetries in ordered phases and applications to quantum disordering,” *SciPost Phys.* **17** (2024) 080, [[arXiv:2308.05730](#)].
- [8] [S. D. Pace](#) and [X.-G. Wen](#), “Exact emergent higher-form symmetries in bosonic lattice models,” *Phys. Rev. B* **108** (2023) 195147, [[arXiv:2301.05261](#)].
- [7] [Y.-T. Oh](#), [S. D. Pace](#), [J. H. Han](#), [Y. You](#), and [H.-Y. Lee](#), “Aspects of $\mathbb{Z}_N$ rank-2 gauge theory in $(2+1)$ dimensions: Construction schemes, holonomies, and sublattice one-form symmetries,” *Phys. Rev. B* **107** (2023) 155151, [[arXiv:2301.04706](#)].
- [6] [S. D. Pace](#) and [X.-G. Wen](#), “Emergent higher-symmetry protected topological orders in the confined phase of $U(1)$ gauge theory,” *Phys. Rev. B* **107** (2023) 075112, [[arXiv:2207.03544](#)].
- [5] [S. D. Pace](#) and [X.-G. Wen](#), “Position-dependent excitations and UV/IR mixing in the $\mathbb{Z}_N$ rank-2 toric code and its low-energy effective field theory,” *Phys. Rev. B* **106** (2022) 045145, [[arXiv:2204.07111](#)].
- [4] [S. D. Pace](#), [C. Castelnovo](#), and [C. R. Laumann](#), “Dynamical Axions in $U(1)$ Quantum Spin Liquids,” *Phys. Rev. Lett.* **130** (2023) 076701, [[arXiv:2109.06890](#)].
- [3] [S. D. Pace](#), [S. C. Morampudi](#), [R. Moessner](#), and [C. R. Laumann](#), “Emergent Fine Structure Constant of Quantum Spin Ice Is Large,” *Phys. Rev. Lett.* **127** (2021) 117205, [[arXiv:2009.04499](#)].
- [2] [S. D. Pace](#), [K. A. Reiss](#), and [D. K. Campbell](#), “The β Fermi-Pasta-Ulam-Tsingou Recurrence Problem,” *Chaos* **29** (2019) 113107, [[arXiv:1908.00564](#)].
- [1] [S. D. Pace](#) and [D. K. Campbell](#), “Behavior and breakdown of higher-order Fermi-Pasta-Ulam-Tsingou recurrences,” *Chaos* **29** (2019) 023132, [[arXiv:1811.00663](#)].

Invited Talks

Colloquia

University of Minnesota [[Slides](#)] Apr '26

Seminars

University of Minnesota [[Slides](#)] May '26

Institute for Advanced Study [[Notes](#)] Jan '26

University of Oxford [[Notes](#)] Nov '25

Simons Center for Geometry and Physics [[Notes](#)], [[Recording](#)] Oct '25

CU Boulder CTQM [[Slides](#)] Sept '25

Georgia Tech [[Slides](#)] May '25

UCLA [[Pre-talk notes](#)], [[Main talk slides](#)] Feb '25

Symmetry Seminar [[Slides](#)], [[Recording](#)] Feb '25

Perimeter Institute for Theoretical Physics [[Slides](#)], [[Recording](#)] Nov '24

Ohio State University [[Slides](#)] Oct '24

Harvard [[Slides](#)] Oct '24

Boston University [[Notes](#)] May '24

Symmetry Seminar [[Slides](#)], [[Recording](#)] Sept '23

Boston University [[Slides](#)] June '22

Max Planck Institute for the Physics of Complex Systems [[Slides](#)] Nov '20

Workshops/conferences

Leinweber workshop on Quantum Fields, Matter, and their Symmetries
[[Slides to appear](#)] Dec '26

Simons Collaboration Workshop on Confinement [[Notes](#)] May '26

OIST TSVP Symposium: Aspects of Generalized Symmetries [[Slides](#)] June '25

OIST Thematic Program: Generalized Symmetries in Quantum Matter
[[Pre-talk notes](#)], [[Main talk slides](#)] June '25

KITP Program: Generalized Symmetries in Quantum Field Theory: High
Energy Physics, Condensed Matter, and Quantum Gravity [[Slides](#)], [[Recording](#)] Apr '25

IBS PCS Workshop: Effective Field Theory Beyond Ordinary Symmetries
[[Slides](#)], [[Recording](#)] Dec '24

SCGP Workshop: Applications of Generalized Symmetries and Topological
Defects to Quantum Matter [[Slides](#)], [[Recording](#)] Sept '24

Teaching Experience

Schools and workshops

Invited TA: [The Physics and Mathematics of Boundaries, Impurities, and Defects](#) Fall '25
[[Lecture 1 recording](#)], [[Lecture 2 recording](#)], [[Notes](#)]

Invited TA: [Atlantic TQFT Spring School 2025](#) Spring '25

Massachusetts Institute of Technology

TA: 8.02, Physics II Spring '26

Two-time guest lecturer: 8.513, Modern Quantum Many-Body Physics Fall '23

Two-time guest lecturer: 8.231, Physics of Solids I Fall '22

Boston University

Undergraduate Teaching Assistant

PY406, Electromagnetic Fields and Waves II Spring '20

PY405, Electromagnetic Fields and Waves I Fall '19

PY452, Quantum Physics II Fall '19

PY451, Quantum Physics I Spring '19

PY410, Statistical Physics & Thermodynamics Spring '19

PY351, Modern Physics I Fall '18

PY313, Waves and Modern Physics Fall '18

Guest lecturer: PY410, Statistical Physics & Thermodynamics Spring '19

Academic Services

Organizing

[Symmetry seminar](#): Organizer 2025 – Present

MIT Ultra Quantum Matter seminar: Organizer 2022 – 2026

MIT physics colloquium committee: Graduate student representative 2021 – 2024

Mentorship

Mentor for [Project SHORT](#) 2020 – Present

MIT UROP Supervisor 2022 – 2023

Mentor for [Boston University's PRISM](#) 2018 – 2020

Journal Referee: SciPost and Physical Review